Name:……Reg No:…….

P525/1
CHEMISTRY
PAPER 1
April-2017
$2\frac{3}{4}$ hours

MWALIMU EXAMINATIONS BUREAU

End of Term 1 Examinations-2017

CHEMISTRY
Paper 1
2 Hours 45 minutes

Instructions to Candidates

*Answer **all** questions in section **A** and **six** questions in section **B**.*

All questions must be answered in the spaces provided.

The Periodic Table, with relative atomic masses, is supplied.

Mathematical tables (3-figure table) are adequate or non-programmable scientific electronic calculators may be used.

Illustrate your answers with equations where applicable.

Where necessary, use the following:

Molar gas constant, R = 8.31 $JK^{-1}mol^{-1}$.

Molar volume of gas at s.t.p is 22.4 litres.

Standard temperature = 273 K

Standard pressure = 101325 Nm^{-2}

For Examiners' use Only														
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Section A: (46 marks)

Answer **all** questions from this section

1. (a) Define the term **standard enthalpy of formation**. (01 mk)

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(b) The standard enthalpies of formation, kJ mol^{-1}, of methane, carbon dioxide and water are -76, -394 and -286 respectively. Determine the enthalpy of combustion of methane. (03 mks)

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2. (a) State the oxidation state of manganese in

(i) MnO_4^- (0½ mk)

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(ii) MnO_4^{2-} (0½ mk)

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(iii) MnO_2 (0½ mk)

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(b) Write equation and state what would be observed when

(i) manganese(IV) oxide is added to hot concentrated hydrochloric acid. (02 mks)

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(ii) manganate(VI) ion is hydrolysed by water. (02 marks)

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3. Complete the following equations and in each case write the accepted mechanism for the reaction

(a) $CH_3CH_2Br + alc\ KOH \xrightarrow{heat}$..... (03 marks)

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(b) $+ Br_2 \xrightarrow[heat]{Fe}$ ..... (03 marks)

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4. (a) Complete the following equations

(i) $^{233}_{91}Pa \rightarrow \beta + \ldots\ldots\ldots\ldots \rightarrow {}^{4}_{2}He + \ldots\ldots\ldots\ldots\ldots$ (02 marks)

(ii) $^{239}_{94}Pu \rightarrow {}^{91}_{39}Y + {}^{146}_{55}Cs + \ldots\ldots\ldots\ldots\ldots$ (01 mark)

(b) A radioactive isotope has a half life of 20 minutes. Calculate the percentage of the isotope that remained after 70 minutes. (02 marks)

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5. (a) Write equation for the reactions that occur when chlorine is bubbled through

(i) aqueous potassium bromide.

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(ii) cold dilute sodium hydroxide.

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(b) the data below shows standard electrode potentials for some half cells.

Half cell	$E^{\theta}(V)$
$Fe^{3+}(aq), Fe^{2+}(aq)/Pt$	$+0.771$
$Cl_2(g), Cl^-(aq)/Pt$	$+1.360$

(i) Is the reaction between iron(II) sulphate and chlorine feasible? Give a reason for your answer (2½ mks)

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(ii) State what is observed when chlorine is bubbled through iron(II) sulphate solution.

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6. Name the reagent that can be used to distinguish between the following pairs of compounds and state what would be observed in each case if each of the member of the pair is separately treated with the reagent you have named.

(a) CH_3OH *and* CH_3CH_2OH (03 marks)

Reagent

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Observations

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(b) $CH_3C \equiv CCH_3$ *and* $HC \equiv CCH_2CH_3$ (03 marks)

Reagent

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Observations

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7. (a) Explain the term '**partition coefficient**' (02 marks)

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(b) 0.5 mole of organic compound was shaken with 60 cm^3 of water and 30 cm^3 of benzene. The partition coefficient between water and benzene is 0.4. Calculate the number of moles of the compound in the aqueous layer. (1½ marks)

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8. (a) Compare the reactivity of the following elements with water.

(i) Magnesium. (02 marks)

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(ii) Barium. (1½ marks)

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(b) Write equation for the reaction between water and the carbide of

(02 marks)

(i) beryllium.

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(ii) calcium.

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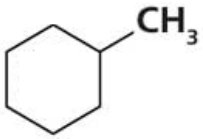
9. Write equation to show how the following compounds can be synthesized.

(a) 

from benzene (3½ marks)

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(b) CH_3COCH_3 from propene (03 marks)

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SECTION B: (54 marks)

Answer **six** questions from this section. Additional questions will **not** be marked.

10. When 3.7 g of an organic compound Q was exploded with excess oxygen, 4.5 g of water and 6.48 dm^3 gaseous mixture were formed. On bubbling the gaseous mixture into a wash bottle containing concentrated potassium hydroxide solution, the volume reduced to 2.0 dm^3. All volumes were measured at s.t.p.

(a) (i) Calculate the empirical formula of Q. (3marks)

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(ii) Determine the molecular formula of Q if its molecular mass is 74. (1 mark)

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(b) Write the structural formulae of all isomers of Q. (2 marks)

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(c) When one of the isomers in (b) above was reacted with a mixture of aqueous iodine solution and sodium hydroxide solution, compound P was formed. When compound P was acidified, compound R was formed. R reacted with phosphorus pentachloride to form compound Z and hydrogen chloride. Identify: (3 marks)

P……..……

R……..……

Z……..……

11. State what would be observed and write equation for the reaction that would take place when

(a) potassium iodide solution is added to acidified potassium dichromate(VII) solution. (2½ marks)

Observation

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Equation

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(b) cyclohexene is added to bromine water. (02 marks)

Observation

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Equation

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(c) acidified solution of hydrogen peroxide is added to iron(II) sulphate solution.

(2½ marks)

Observation

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Equation

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(d) copper turnings added to hot concentrated nitric acid. (02 marks)

Observation

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Equation

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12. (a) Explain what is meant by the term **ideal solution**. (02 marks)

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(b) At standard atmospheric pressure, hydrochloric acid and water form a constant boiling point mixture having a boiling point of $110^{o}C$ and composition of 20% by mass of hydrochloric acid.

(i) Define the term **constant boiling point mixture**. (01 mark)

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(ii) Sketch a labeled diagram of the boiling point-composition for hydrochloric acid and water system. [The boiling points of water and hydrochloric acid are 100^oC and 85^oC respectively] (03 marks)

(iii) Describe what would happen if a mixture of 10% hydrochloric acid is fractionally distilled. (2½ marks)

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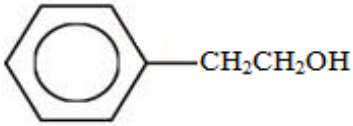
13. Complete the following equations and in each case write the accepted mechanism for the reaction

(a) 

$\xrightarrow[180^oC]{Conc\ H_2SO_4}$ ..... (03 marks)

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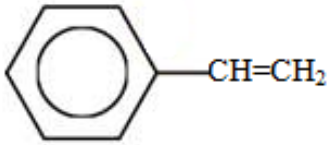
(b) 

$\xrightarrow[H_2O, warm]{conc\ H_2SO_4}$ ..... (03 marks)

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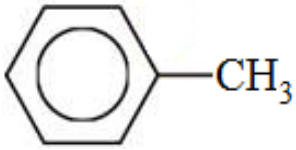
(c) 

$+\ Cl_2 \xrightarrow[light]{uv}$ .....(03 marks)

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14. Although lithium is in Group I of the Periodic Table, it resembles elements of Group II in particular magnesium in some of its properties.

(a) (i) State **four** properties in which lithium and magnesium are similar.

(04 mks)

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(ii) Give **one** reason for the chemical similarity between the two elements. (01 mk)

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(iii) Name **one** other pair of elements that show similarity like lithium and magnesium..

(01 mk)

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(b) Write equation

(i) to show how anhydrous magnesium chloride can be prepared.

(1½ mks)

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(ii) for the reaction that occurs when hydrated magnesium chloride, $MgCl_2.6H_2O$ is heated gently. (1½ mks)

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15.(a) Write

(i) equation for the hydrolysis of sodium benzoate in water. (01 mk)

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(ii) the expression for the hydrolysis for the hydrolysis constant, K_h of sodium benzoate. (01 mk)

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(b) A solution contains 24.2 g of benzoate ions per litre at 25°C. Calculate the pH of the solution. [The hydrolysis constant of sodium benzoate is $1.6 \times 10^{-10}\ mold\ dm^{-3}$; ionic product of water is$10^{-14}\ mol^2 d\ dm^{-6}$ at 25°C]. (04 mks)

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(c) Explain what would happen to the pH of a solution containing benzoate ions and benzoic acid, if to it was added a small amount of sodium hydroxide solution. (03 mks)

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16. Explain each of the following observations.

(a) When dilute ammonia solution is added to a solution containing aluminium ions, a white precipitate is formed however similar treatment of the solution of aluminium ions in the presence of ammonium ions no precipitate is formed. (5 mks)

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(b) The melting points of group II elements are much higher than those of the corresponding group I elements. (4 mks)

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17. Name **two** major sources of sulphur dioxide for use in the manufacture of sulphuric acid. (01 mk)

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(b) Sulphur dioxide reacts with oxygen to form sulphur trioxide according to the following equation.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \qquad \Delta H^{\theta}_{298\,K} = -197\ kJ\ mol^{-1}$

State what would happen to the position of the equilibrium and in each case give a reason if,

(i) the pressure of the system is increased. (1½ mks)

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(ii) the temperature of the system is increased. (1½ mks)

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(iii) vanadium(V) oxide is added to the system. (1½ mks)

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(c) When 1.4 moles of sulphur dioxide and 0.8 moles of oxygen were placed

in a 2 litres vessel and heated at 700°C, 1 mole of sulphur trioxide was formed at equilibrium. Calculate

(i) the amount of sulphur dioxide and oxygen at equilibrium in moles per litre. (1½ mks)

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(ii) the equilibrium constant, K_c for the reaction and state the units (2 mks)

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End